Quantum Dynamics Project – LTTC Winter School

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In this study, the propagate.f90 program from the QD tutorial is modified. Instead of propagating a Gaussian wave packet, the code now propagates a wave packet constructed as a superposition of quantum harmonic oscillator eigenstates. This is achieved by modifying the initpsi subroutine to compute the eigenstates using the harmonic oscillator formula and Hermite polynomials.

The completed tasks are explained in this report, and the corresponding files have been added to the homework folder.

Task1: Subroutine hermite_poly that computes Hermite polynomials $H_n(y)$ using the recursion (1) is added to the propagate.f90 code. Loop is created from n = 2 to nmax.

$$H_n(y) = 2\,y\,H_{n-1}(y) - 2(n-1)H_{n-2}(y) \qquad (1)$$

Subroutine takes an input value y and an integer nmax; and base cases (2) are defined in the subroutine.

$$H_0(y) = 1\ ,\ H_1(y) = 2y \qquad (2)$$

Task2: Subroutine initpsi is modified by adding a loop to the code to calculate the eigenstates $\phi_n(x)$ defined in (3).

$$\phi_n(x) = \frac{1}{\sqrt{2^n\,n!}}\left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \exp\left[-\frac{m\omega}{2\hbar}x^2\right] H_n\left(\sqrt{\frac{m\omega}{\hbar}}\,x\right) \qquad (3)$$

The code uses the provided harmonic oscillator formula (via hermite_poly subroutine) for the Hermite polynomial recursion to compute $\phi_n(x)$ for n= 0 to 4.

Note: In the code reduced Planck's constant ($\hbar$) is set to 1.

Task 3: Subroutine initpsi is modified so that it calculates the wave packet in (4) by truncating the sum at n=4.

$$\Psi_0(x) = \sum_{n=0}^{4} c(n)\,\phi_n(x) \qquad (4)$$

Last version of the code constructs a wave packet by reading coefficients from the coefficients file and builds the initial wavefunction as a superposition.

Task 4: Wave packet is normalized. Normalization condition is shown in (5).

$$\int |\Psi_0(x)|^2 dx = 1 \qquad (5)$$

In the end of the initpsi subroutine, normalization block calculates the "norm" via discrete integration and scales $\Psi_0(x)$ so that its total probability is 1. This ensures that the wavefunction is normalized.

Task5: To debug the program, the coefficients are set up such that only one coefficient is 1 and all others are 0, and the initial wave packet is positioned at the bottom of the potential. Under these conditions, the initial wave packet corresponds to a pure eigenstate of the harmonic oscillator. When the propagation is performed using the split-operator method, observations from the generated harmonic5.gif file confirm that the spatial profile remains invariant throughout the evolution, showing that only a phase factor is acquired over time.

This behavior verifies that the code correctly propagates a pure eigenstate without altering its probability density, as expected.

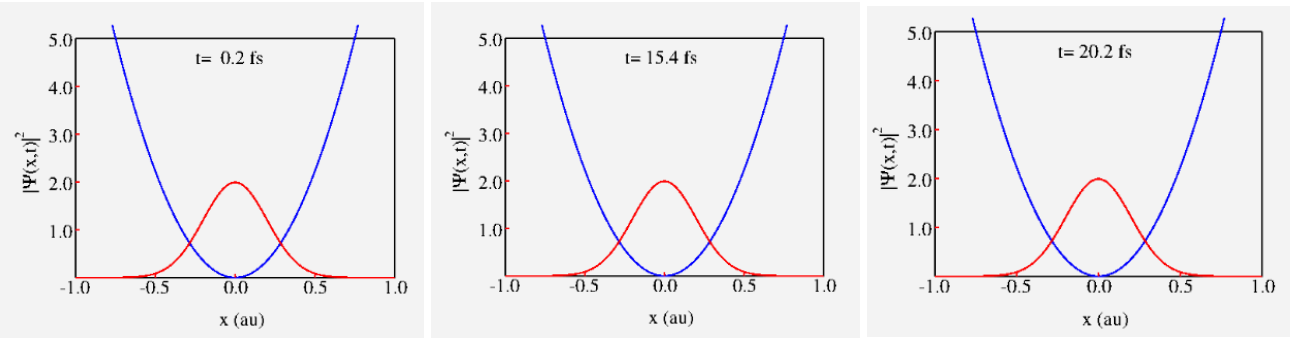

Figure 1. Snapshots at different times during the propagation of the wavepacket from the harmonic5.gif file.

Task6: Following three cases when coefficients are set to different values(by modifying the coefficients file) are discussed.

Case 1: All Coefficients Equal to 1

When all five coefficients are set to one, the initial wave packet is a superposition of the first five harmonic oscillator eigenstates without any imposed symmetry. As a result, the probability density exhibits multiple peaks and intricate interference patterns that evolve over time, reflecting the complex interplay of the different phase factors. The overall period T is determined by the composite oscillations of all eigenstates, leading to a beating pattern that does not correspond to a single, simple frequency.

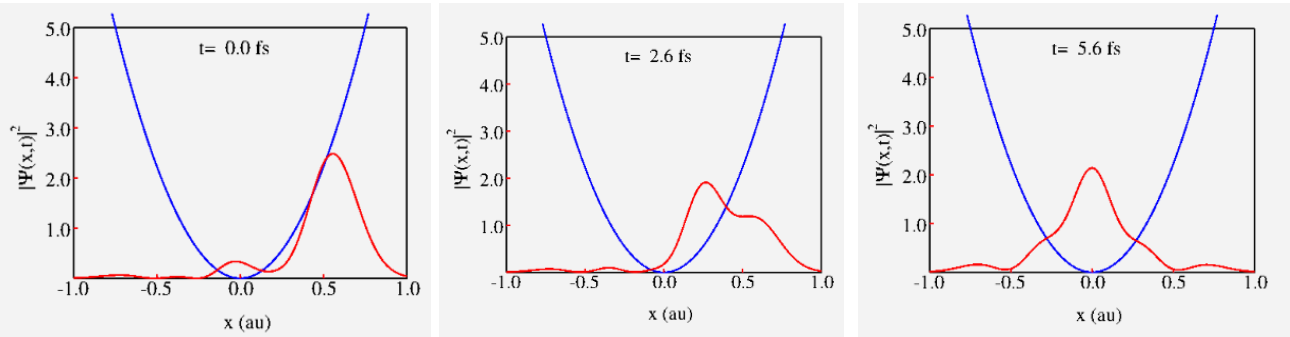

Figure 2. Snapshots at different times during the propagation of the wavepacket from the harmonic6-1.gif file.

Case 2: Only Even Coefficients (c0, c2, c4) Equal to 1

By activating only the even coefficients, the resulting wave packet becomes an even function, therefore it is symmetric about x=0. In this configuration, the probability density remains symmetric throughout the propagation, as the contributions from the even eigenstates reinforce the same shape on both sides of the origin. The period T in this case is more regular, being directly linked to the energy differences between the even states, which results in a clear and predictable oscillatory behavior.

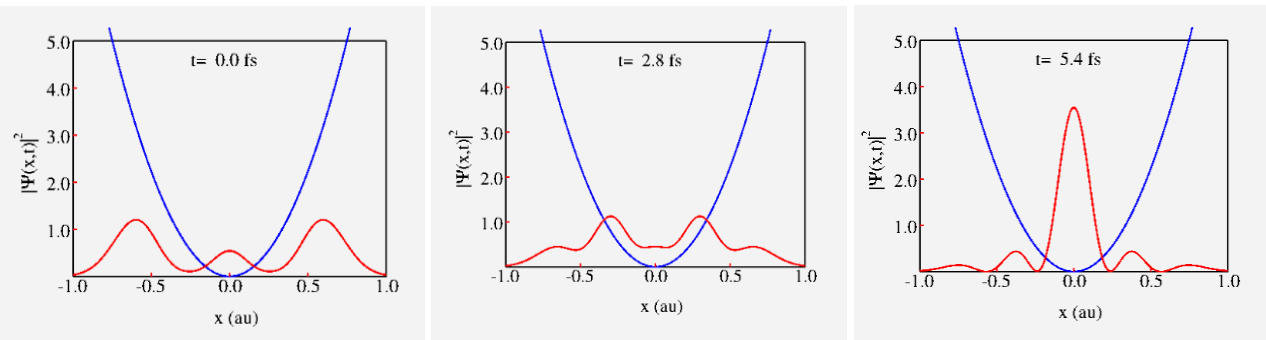

Figure 3. Snapshots at different times during the propagation of the wavepacket from the harmonic6-2.gif file.

Case 3: Only Odd Coefficients (c1 and c3) Equal to 1

With only the odd coefficients nonzero, the constructed wave packet is an odd function and therefore antisymmetric about x=0. This antisymmetry is evident in the probability density, and there is a distinct node at x=0. Despite the antisymmetric nature of the wavefunction, the probability density appears symmetric, but its evolution is governed by the phase difference between the odd eigenstates. Consequently, the period T is set by the energy gap between the n=1 and n=3 states, leading to an oscillatory behavior that is uniquely different from the cases with even or mixed coefficients.

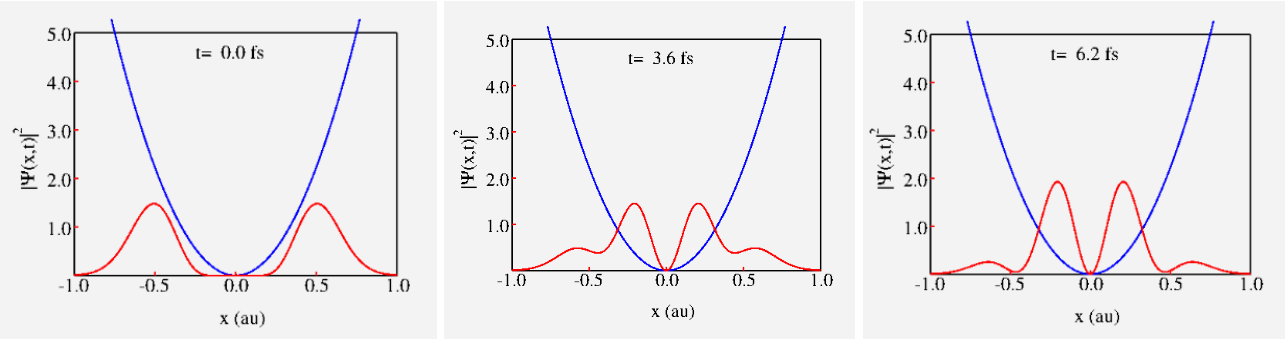

Figure 4. Snapshots at different times during the propagation of the wavepacket from the harmonic6-3.gif file.